Progressive exercises

E1 · Direct application. Let X a random variable such that $E(X) = 3$ and $V(X) = 2$. Set $Y = 4X - 1$. Calculate the expectation, variance and standard deviation of Y.

E2 · Application and justification. Random variables X and Y are independent and satisfy $E(X) = 2$, $E(Y) = 5$, $V(X) = 1$ and $V(Y) = 4$. Calculate $E(2X - Y)$ and $V(2X - Y)$.

E3 · Method to organise. Consider 64 independent random variables with the same distribution, with expectation 10 and standard deviation 4. Calculate the expectation and standard deviation of their mean.

E4 · Choice of strategy. Let X a random variable such that $E(X) = 100$ and $V(X) = 9$. Using the Bienaymé-Chebyshev inequality, give an upper bound for $P(|X - 100| \geq 6)$.

E5 · Synthesis or proof. Let M_n the mean of n independent random variables of same distribution, with expectation μ and variance 4. Determine a size n sufficient to ensure $P(|M_n - \mu| < 0{,}2) \geq 0{,}95$.